Ghana's Cities Are experiencing dumsor despite sitting on a sun mine

Apart from death, there is no word that any Ghanaian dread the most than “dumsor.” The word, which is an amalgamation of “dum” (off) and sor (on), has come to dominate Ghana’s public discourse. Decades of load-shedding have ingrained itself into the social fabric of every Ghanaian society. There is a running joke that if you run out of excuse to give as to why you haven’t performed a task, you can just say that you are experiencing dumsor and no one would ever doubt you. Ghanaians have come to accept that they will never experience stable power and it is a shame that we have come to accept such a social reality when we are literally sitting on a sun mine. Research that we conducted using data from PVGIS SARAH3 and the Ghana Statistical Service indicates that Ghana has the solar energy potential to cover its residential electricity demand many times over. We analyzed the situation by looking at five populous city, Accra, Kumasi, Takoradi, Tamale and Tamale. The results from the data paints a compelling picture; Ghana should not be suffering from dumsor when we have an untapped potential, rooftop solar photovoltaic (PV). Ghana’s urban rooftops can be a source of untapped energy in solving our power crisis.

The Numbers That Should Shake Energy Policy

We modelled this scenario, “what would happen if each housing unit in these aforementioned cities had a standard 5 kilowatt-peak (kWp) rooftop solar system installed?” The metric used to judge adequacy was simple: “could the energy generated meet assumed residential demand of 500 kWh per person per year? Our statistical model across these five cities gave a resounding yes. We calculated a self-sufficiency ratio for each city; a score above 1.0 means rooftop solar alone was enough to cover all the residential electricity needs. Here is what we found; Accra had a 9.2 times self-sufficiency ratio, Kumasi had a 5.8 times self-sufficiency ratio, Tema had a 6.4 times self-sufficiency ratio, Tamale and Takoradi had a 5.5- and 5.6-times self-sufficiency ratio. Accra’s wide margin is not as a result of most sunshine per square meter but because of its large population and large housing units. With a rough estimate of 558 housing unit per 1,000 people (70% higher than Tamale’s 323), the capital city has the largest aggregate surface area for solar panels, giving it a total solar potential of a total solar potential of approximately 13,556 kW/m² — about 52% above the five-city average.

Where the Sun Shines Hardest

Based on our model, we found out that with regards to raw solar intensity, the north has the edge. Tamale is sitting in Ghana's upper belt and largely outside the heavy cloud of the forest zone. Using a 20-year historic data (2005 – 2025), the city has consistently recorded the highest annual energy output per system, with a median of approximately 8,775 kWh per year. By contrasts, Kumasi sits on the rainforest belt and records the lowest median of approximately 7,980 kWh per year. Despite the incidence of climate change witnessed in the world, our data shows that in the last 20 years, there has not been a long-term decline in solar resources and this is a positive sign that investment in solar will still pay off in the future. Below is a snapshot of the cities compare on key metrics.

City	Per-Capita Solar (kWh/yr)	Self-Sufficiency
Accra	~4,600	9.2×
Tema	~3,150	6.4×
Kumasi	~2,875	5.8×
Tamale	~2,850	5.7×
Takoradi	~2,800	5.6×

So Why Aren't We Already Doing This?

It seems that the simplistic answer to our energy crisis is to nationwide solar adaptation but then the situation on the ground tells a different picture. On the demand side, many households cannot afford solar PV. There is the little awareness of the technology's benefit to individual residents both in rural and urban areas. Aside this, there is the general assumption that electricity is the government's job and not that of the households. On the supply side, the country has too few certified solar installer and no reliable equipment quality standards. Ghana has a Renewable Energy Act 832 of 2011 and this introduced feed-in tariff mechanisms. However, like most implementation in Ghana, the rollout of the feed-in tariff mechanism has been slow and this undermines investor confidence. Grid connection procedures are still opaque and the financial instruments needed to make solar accessible is still absent. Interest rate are still exorbitant despite the Bank of Ghana's attempt to make loans cheaper. A cheaper and more accessible financial instrument for solar will go a long way to make solar PV accessible to the masses.

What Needs to Happen Next to drive solar PV adoption?

The country needs to:

1. Expand and properly enforce net-metering and feed-in tariff mechanisms under the existing Renewable Energy Act.
2. Develop accessible financing instruments such as solar leasing schemes, green bonds, and SSNIT-backed pension-linked solar loans to reduce the upfront cost barrier for urban households.
3. Invest in grid modernisation in high-density cities like Accra and Tema to handle two-way power flows as distributed generation grows.
4. Prioritise Tamale for off-grid and mini-grid solar solutions, given its higher irradiance and lower grid reliability.
5. Commission high-resolution building footprint mapping via remote sensing to sharpen estimates of actual usable rooftop area.

We are not promising instant energy independence but we are advocating an energy transition to solar PV as a way of solving the country's persistence energy crisis. However, our recommendations are likely to be impacted by a few things such as shading, structural suitability, informal settlement layouts, and grid capacity all limit how quickly actual deployment can scale. The assumed 5 kWp system size may also overestimate what is feasible on smaller urban dwellings. Despite these shortcomings, the fundamental issue is that the country has the solar resources and our cities are not utilizing our solar potential energy. We tend to assume that we are resource poor but this is not the issue when it comes to electricity. We are only policy-poor and finance-poor.

The Bottom Line

Ghana has spent decades managing an electricity crisis while sitting beneath one of the most abundant solar resources on the continent. Accra alone has a per-capita solar energy availability of roughly 4,600 kWh per person per year more than nine times the assumed residential demand. Every city in this study exceeds that demand threshold by a factor of at least five. Rooftop solar is not a niche solution for eco-conscious households. In the Ghanaian context, it is a viable, data-backed pathway to urban energy resilience. The question is no longer whether the sun can power our cities. It is whether our policies, financial systems, and institutions will finally let it.